

How does national SOC monitoring on agricultural soils align with the EU strategies? An example using five case studies

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ABSTRACT ORIGINAL

Soil functioning contributes to the delivery of a vast range of ecosystem goods and services, and ecosystem health is therefore reflected by the capacity of the soil to perform underlying functions. Soil organic carbon (SOC) is a key indicator for soil quality as it is an integral driver of many soil functions and associated ecosystem services. Across the globe, SOC stocks are declining due to expanding agriculture and unsustainable practices. Awareness of the fact that soil is a non-renewable resource and its functioning important for all life on Earth is increasing, especially among policymakers. As such, goals for the preservation and restoration of SOC are formulated in policies under the European Green Deal. However, the evaluation of these goals at the European level is hampered by a non-harmonized diversity in national SOC monitoring strategies. While some SOC indicators can be useful for the evaluation of most policy goals (i.e., baseline and potential SOC stocks), additional and contrasting SOC data are often required for the evaluation of the goals formulated by the different EU directives. This study provides an overview of five ongoing SOC monitoring programmes across Europe and discusses how national programmes may be aligned to evaluate goals at the EU level. Five countries with very different soil monitoring programmes were included in a case study to illustrate the potential for harmonization and standardization of SOC assessment. Based on this study, we conclude that SOC monitoring strategies can be harmonized, but not standardized. We further suggest five sampling strategies that have potential for harmonization under the proposed Directive on Soil Monitoring and Resilience. © 2024 The Authors. European Journal of Soil Science published by John Wiley & Sons Ltd on behalf of British Society of Soil Science.